Practical No : 9

Aim: Develop an Android application implementing SQLite database CRUD operations.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity"
  android:orientation="vertical">

  <EditText
    android:id="@+id/id"
    android:hint="ID"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

  <EditText
    android:id="@+id/name"
    android:hint="Name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
  <EditText
    android:id="@+id/lname"
    android:hint="Enter Last Name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

  <Button
    android:id="@+id/insert"
    android:text="INSERT"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

  <Button
    android:id="@+id/view"
    android:text="VIEW"
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"/>

  <Button
    android:id="@+id/update"
    android:text="UPDATE"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

  <Button
    android:id="@+id/delete"
    android:text="DELETE"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

</LinearLayout>
```

DBHelper.java

```
package com.example.database;

import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

  public DBHelper(Context c) {
    super(c, "StudentDB", null, 1);
  }

  public void onCreate(SQLiteDatabase db) {
    db.execSQL("CREATE TABLE student(" +
        "id INTEGER PRIMARY KEY, name TEXT, lname TEXT)");
  }

  public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
    db.execSQL("DROP TABLE IF EXISTS student");
    onCreate(db);
  }

  void insert(int id, String name, String lname) {
    getWritableDatabase().execSQL(
        "INSERT INTO student VALUES(" + id + ",'" +
```

```
                name + "','" + lname + "')");
  }

  void update(int id, String name, String lname) {
    getWritableDatabase().execSQL(
        "UPDATE student SET name='" + name +
            "', lname='" + lname + "' WHERE id=" + id);
  }

  void delete(int id) {
    getWritableDatabase().execSQL(
        "DELETE FROM student WHERE id=" + id);
  }

  Cursor view() {
    return getReadableDatabase().rawQuery(
        "SELECT * FROM student", null);
  }
}
```

MainActivity.java

```
package com.example.database;

import android.database.Cursor;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
  EditText id, name,lname;
  Button b1,b2,b3,b4;
  DBHelper db;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
```

```
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
id = findViewById(R.id.id);
name = findViewById(R.id.name);
lname = findViewById(R.id.lname);
b1=findViewById(R.id.insert);
b2=findViewById(R.id.update);
b3=findViewById(R.id.delete);
b4=findViewById(R.id.view);
db = new DBHelper(this);

b1.setOnClickListener(v ->
        db.insert(
                Integer.parseInt(id.getText().toString()),
                name.getText().toString(),
                lname.getText().toString()));

b2.setOnClickListener(v ->
        db.update(
                Integer.parseInt(id.getText().toString()),
                name.getText().toString(),
                lname.getText().toString()));

b3.setOnClickListener(v ->
        db.delete(
                Integer.parseInt(id.getText().toString())));

b4.setOnClickListener(v -> {

    Cursor c = db.view();
    String data = "";

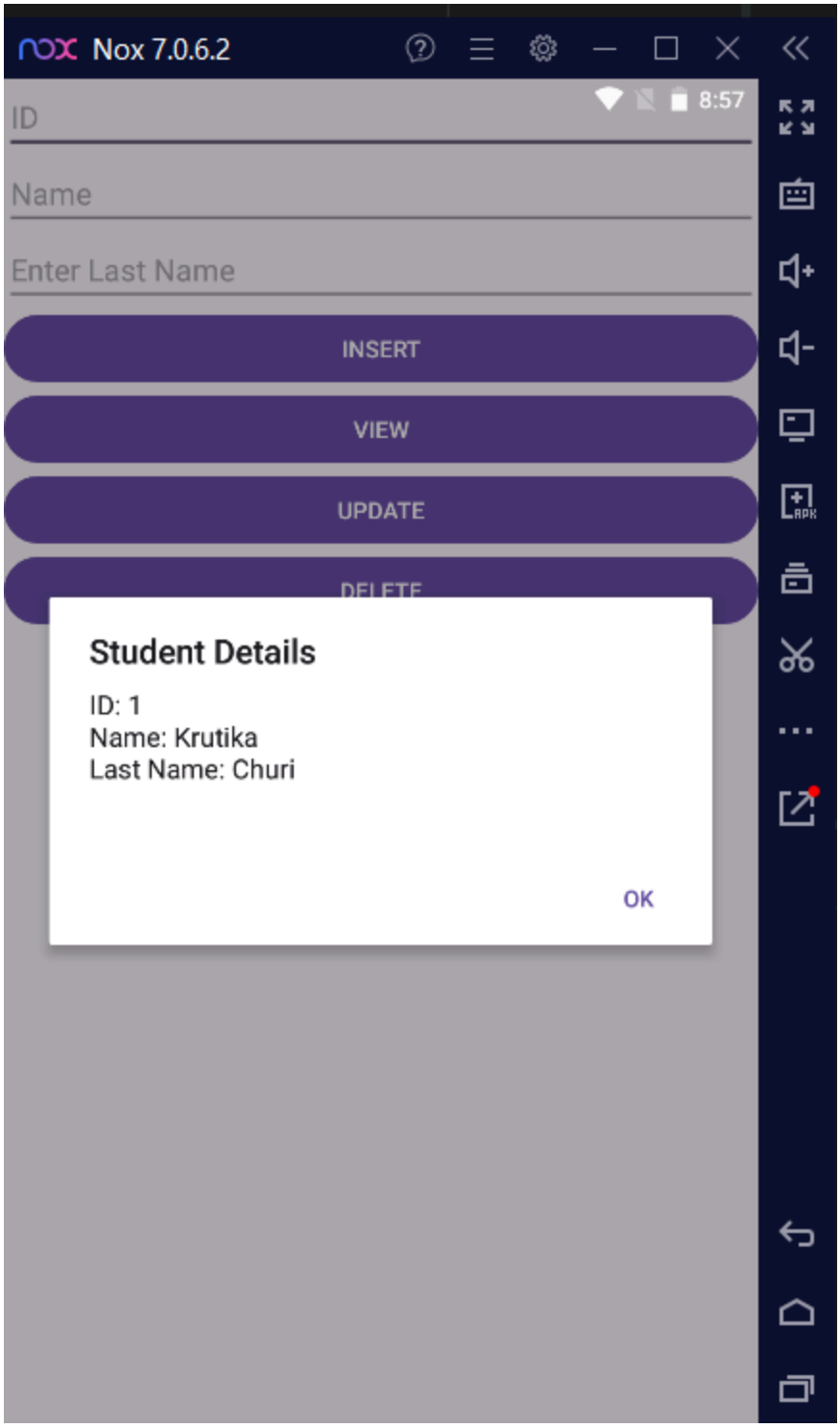
    while (c.moveToNext()) {
        data += "ID: " + c.getInt(0) +
                "\nName: " + c.getString(1) +
                "\nLast Name: " + c.getString(2) +
                "\n\n";
    }

    new AlertDialog.Builder(this)
            .setTitle("Student Details")
            .setMessage(data)
            .setPositiveButton("OK", null)
            .show();
```

By: Prof. Krutika H. Churi
SDSM College, Palghar

```
      });
  }
}
```

Output:



By: Prof. Krutika H. Churi
SDSM College, Palghar